Creation Nation Square Sync

Feature Guide: everything the plugin can do

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Creation Nation Square Sync brings the Square categories **you choose** onto a WooCommerce website, keeps stock matched between the two, and can record website orders in Square so sales and stock live in one place. Square stays the master catalog; the website follows it. It is built to work gently on large catalogs and modest servers.

1. At a glance

Capability	What it does
Selective category sync	Tick Sync or Skip on each Square category (sub-categories included). Only what you choose reaches the website.
Full product data	Names, descriptions, prices, SKUs, barcodes, tax status, categories, photos, simple and variable products.
Two-way stock	Square counts flow to the site instantly by webhook, with timed checks as a safety net. Website sales lower Square stock.
Website orders in Square	Three modes: lower stock only, record a full paid order, or do nothing. Cancellations and refunds are reversed.
Per-product manual sync	A Sync button on the product screen, a row link in the product list, and a bulk action.
Targeted tools	Sync one category, find any Square product by name or SKU and add it, send a specific order to Square.
Keep-hidden memory	Products you hide by hand stay hidden through every future sync.
Automatic SKUs	Products synced by hand that lack a SKU get the next code in a running sequence, written into Square so it stays the master.
Barcode labels	Print Code 128 labels on Dymo 30336 stickers from any product, with quantity controls and printer calibration.
Self-healing	Small paced batches, automatic backoff, checkpoint and resume after crashes, retrying orders, a watchdog.
Token safety	Encrypted token, tested before saving, with an alarm banner, log entry and email if Square rejects it.

2. How the data flows

Square stays the master catalog. Instant updates arrive by webhook; timed checks are the safety net.

Direction	What moves	How it is triggered
Square to website	Categories, products, variations, prices, SKUs, barcodes, photos	Full sync every 6 hours; on demand; about 60 seconds after Square reports a catalog change; per product or per category by hand
Square to website	Stock levels	Instant webhook; 15-minute incremental check; full stock check on demand and after each full sync
Website to Square	Stock reductions, optionally full paid orders	When an order reaches Processing or Completed
Website to Square	Refund and restock	When an order is cancelled or refunded

3. Catalog sync in detail

Products

- **Simple vs variable.** One Square variation becomes a simple product. Several become a variable product with an *Option* attribute. Blank variation names become “Regular”; duplicate names are numbered. If an item changes type in Square, the website product is converted.
- **Prices.** Fixed prices become the regular price. Items with variable pricing sync without a price.
- **SKUs.** Copied from Square. If another website product already uses the SKU, the plugin logs a conflict and leaves the SKU off rather than failing.
- **Barcodes.** Square UPC/GTIN values are saved on the product and, where WooCommerce supports it, in the GTIN field.
- **Tax and description.** Taxable status follows Square. HTML descriptions are sanitised; plain-text descriptions are formatted into paragraphs.
- **Publish or review.** New and updated products arrive Published, or as Drafts for review, as you choose.

Photos

- Downloaded into the media library; first image is the featured image, the rest form the gallery; alt text is the product name.
- Fetched one product at a time, at least 20 seconds apart, and only when a product’s Square image set changes. Photos already downloaded are reused.
- If an item has no photos in Square, the website product’s photos are cleared.

Staying efficient

- Items whose Square version has not changed are skipped, so repeat syncs are cheap.
- Work runs 10 items per step with a 15-second pause between steps, so the server always has breathing room.
- Stock for each batch is set immediately after the batch, so products appear in stock as they arrive.

4. Choosing what goes online

Rule	Behaviour
Sync	Brings the category and all of its sub-categories onto the website as matching product categories (same hierarchy; renames and moves follow).
Skip	Keeps a category and its sub-categories off, even inside a synced category. Skip always wins.
Price-button pattern	An optional regular expression (validated on save, blank by default) that keeps register-only items off the site, for example one matching names such as “CD 1.95”, “LP 9.95” or “Video 4.95”.
Archived or deleted	Never synced. If a product’s item is archived, deleted, or leaves every synced category, the website product is set to Draft. It returns automatically if the item comes back.
No variations	An item with no live variation is not synced.
Multiple categories	An item in several Square categories appears in the matching website categories.

Keep hidden. If you move a synced product from Published to Draft yourself, the plugin remembers and keeps it hidden through every sync, including manual ones. Publishing it again clears the flag. Products hidden by the sync itself (because they left a synced category) are not flagged and reappear when they qualify again.

5. Inventory

- **Square is the source of truth** for stock, read for the one location you choose.
- **Instant updates.** A signed Square webhook (inventory count updated) changes the website’s stock the moment Square’s changes.
- **Incremental check every 15 minutes** catches anything a webhook missed, using the time of the previous check with a short overlap.
- **Full stock check** on demand and after every full catalog sync, in chunks of 150 variations. Anything Square has no count for becomes 0.
- Respects each item’s *track inventory* setting and per-location overrides. Untracked items show In stock, or Out of stock when flagged sold out at that location.
- Quantities are whole numbers: fractions round down and negatives become 0.

6. Website orders

Mode	Use when	What happens
Lower Square stock only (default)	Customers pay through Square	The sale already appears in Square through the payment. Each paid order just reduces Square stock.
Record as a paid Square order	Payments go through Stripe, PayPal or another processor	Each paid order becomes a paid Square order, so it shows in Square sales and item reports and Square lowers its own stock.
Do nothing in Square	You handle it yourself	No order or stock activity is sent.

Paid-order mode in detail

- Catalog items link to their Square items. Website-only items go in as custom lines at their website price.
- Discounts, shipping, and the sales tax collected online are recorded as fixed amounts so Square's total matches what the customer paid. Square's own tax rules are not applied on top.
- The payment is recorded as an external card payment tagged as a website sale and labelled with the site name, with the web order number as its reference. Zero-total orders are recorded with no payment.
- If Square's total differs from the website's, a note on the order states the difference.

Both active modes

- **Reversals.** A cancelled or fully refunded order puts its items back into Square stock and, in paid-order mode, records a refund for the full amount.
- **No duplicates.** Every Square request uses a stable idempotency key that includes a short code unique to the site, so a retry cannot record an order twice and several sites can share one Square account safely.
- **Automatic retries** with growing delays (5 minutes times the attempt number), up to 6 attempts. After that, the order note says what to do.
- **Watchdog.** Every 5 minutes it re-checks paid orders from the last 14 days (20 per pass) that never reached Square.
- **Manual push.** Type a website order number on the admin screen to send it to Square on the spot.
- Outcomes are written to the WooCommerce order notes and to the activity log.

7. Hands-on tools

The Square Sync screen (WooCommerce menu)

Section	Contents
Status	Catalog sync state or live progress; items on the website from Square; instant-updates status with the time of the last Square message; last stock check; last sync summary (updated, hidden, errors).
Settings	Access token, Square location, category tree with Sync and Skip boxes, price-button pattern, new-product status, website order mode.
Tools	Sync now; Check all stock now; Turn on or Reconnect instant updates; Send website order #; Sync one category; Find a product in Square.
Activity	The latest 100 events (300 are kept), with errors highlighted.

Sync one category

Re-pulls every item in a chosen synced category and its sub-categories, whether or not Square says anything changed. It shows live progress, warns if it stalls for 10 minutes, and has a Stop button. Only one runs at a time, and the usual rules (Skip, price buttons, Keep hidden) all apply.

Find a product in Square

Search Square by name or SKU (up to 25 matches). Each result shows its category, price range, variation count, SKUs and its state on the website (Published, Kept hidden, Draft, or Not yet). A button adds it to the website or refreshes it; photos are fetched right away. An item that must stay off shows the plain-words reason: archived, price button, skipped category, not in a synced category, no category, or no variations.

Per-product manual sync

Where	How
Product edit screen	A “Square” box shows link status, time of the last manual sync and any Keep-hidden flag, with a Sync now button.
Products list	A “Sync with Square” link under each linked product.
Bulk action	“Sync with Square” on selected products, up to 25 per click.

After each sync a message lists what changed (name, price old to new, stock old to new, published or hidden, photo) or says the product is already up to date. Photos are fetched immediately rather than queued. If the item no longer belongs on the website, a one-click Hide this product link is offered. Manual sync requires permission to edit the product.

Automatic SKUs

When you sync a single product by hand (product-screen button, row link, or Find a product in Square), any variation with no SKU in Square is given the next code in a running sequence, for example SHOP1001. The prefix is a setting (letters and numbers, up to 10); leave it blank and automatic SKUs are off.

- Codes are one above the highest ever issued, are never reused, and are checked against SKUs already on the website.
- The code is written **into Square**, so Square remains the master and the sync copies it back. The result message shows the new SKU.
- The Tools area displays the next code to be issued, so codes typed by hand can follow the same sequence.
- Only items that belong on the website receive codes. The bulk action does not assign SKUs. This is the one place the plugin writes to your Square catalog, so the token needs catalog write access.

Barcode labels

Prints Code 128 labels from the browser on Dymo 30336 stickers (1 in by 2-1/8 in). Each label carries the product name, a barcode of the SKU (or the UPC when there is no SKU), the shop name, the code and the price. The shop name is a setting and defaults to the site name.

Feature	Detail
Where to start	A Barcode label box on the product screen, a Print label row link, or the Print barcode labels bulk action (up to 200 products per screen).
Label screen	A live preview of every label; quantity per product (up to 500); Set all to 1; Set all to stock on hand; a running label total.
Variations	Variable products print one label per variation. Items with no SKU or UPC are flagged and skipped.
Calibration	Nudge the print position right and down, up to half an inch each way, if labels land off-centre. Saved for next time.
How it prints	Through the browser to the Dymo printer. Barcodes are drawn with JsBarcode, loaded from cdnjs on the label screen only.

8. Reliability and self-healing

Situation	What the plugin does
A step takes over 45 seconds	Pauses syncing for 3 minutes (10 minutes if a step took over 2 minutes) and logs why.
The server is busy or paused	Retries the same batch every 2 minutes until it is clear.
The server crashes mid-sync	Progress is checkpointed after each batch. The watchdog (every 5 minutes) resumes from the checkpoint once it has been idle for 10 minutes.
A lock gets stuck	Cleared automatically after 10 minutes and the sync restarts. Locks older than 30 minutes are treated as stale.
A sync is requested while one runs	It is flagged and starts right after the current one finishes. Nothing runs in parallel.
Square or the network hiccups	Each call is tried up to 3 times (network errors, rate limits, 5xx responses) with waits of about 1 and 3 seconds plus random jitter, 45-second timeout.
Square rejects the token (401)	Admin-wide error banner, log entry, and one email per outage. Clears itself once a working token is saved.
A website order fails to reach Square	Automatic retries (up to 6), then the watchdog, then a manual Send to Square.

9. Security

- **Token encrypted at rest** with AES-256-CBC using a key derived from the site's WordPress salts. It is never displayed again after saving.
- **Token tested before saving.** A new token is verified against Square first; if rejected, the previous one is restored.
- **Signed webhooks only.** Every webhook is checked with HMAC-SHA256 against Square's signature key; mismatches get a 403. The signature key is also stored encrypted.
- **Permissions.** Admin screen and actions require the manage-WooCommerce capability; per-product actions require permission to edit that product. All actions use nonces.
- **Clean disconnect.** Disconnecting unregisters the webhook and removes the token.
- No credentials are written into the plugin's code.

10. Settings reference

Setting	Meaning
Access token	Production token from your Square developer app. Leave blank to keep the saved one; tick Disconnect to remove it.
Square location	The location whose stock counts and sold-out flags drive the website.
Categories: Sync / Skip	Which Square categories (with sub-categories) go online and which stay off. Skip wins.
Price-button pattern	Optional regular expression; matching item names stay off the website. Blank skips nothing. Invalid patterns are rejected.
New and updated products are	Published, or Saved as drafts for review.
Website orders	Lower Square stock only; Record as a paid Square order; Do nothing in Square.
Automatic SKU prefix	Letters and numbers, up to 10. Variations with no SKU get the next code with this prefix when synced by hand. Blank turns automatic SKUs off.
Shop name on labels	Printed on barcode labels. Blank uses the site name.
Pause switch	Setting the dzsq_paused option makes the plugin stay completely idle (no screen setting).

11. Schedules and timings

Job	Timing
Incremental stock check	Every 15 minutes
Watchdog	Every 5 minutes
Full catalog sync	Every 6 hours; on demand; about 60 seconds after a Square catalog-change webhook
Catalog batch	10 items per step; 15-second pause between steps
Category sync batch	10 items per step; 5-second pause between steps
Photo downloads	One product at a time, at least 20 seconds apart
Full stock check	150 variations per chunk; 10-second pause between chunks
Order retries	5 minutes times the attempt number; maximum 6 attempts

12. Good to know

- **Square wins on synced fields.** Name, description, price, SKU, barcode, categories and photos are overwritten whenever an item syncs, so edit them in Square. Website-only products with no Square link are never touched.
- **The catalog flows one way** (Square to website). Price or stock edits made in WooCommerce are not pushed to Square. The two exceptions are stock reduced by website sales and SKUs assigned automatically.
- One Square account and one location per site.
- Talks to Square's production API only; there is no sandbox mode, so test with a small category and new products set to Saved as drafts for review.
- Label printing supports the Dymo 30336 size only.
- Partial refunds are not mirrored; whole-order cancellation and refund are.
- Item modifiers and custom attributes are not synced. Discounts are recorded as a single "Website discount" line.
- Not formally tested with WooCommerce High-Performance Order Storage or multisite. Deleting the plugin removes its settings, saved token, webhook key and activity log but leaves products, categories, media and orders in place; tick Disconnect first so the webhook is also removed from Square.
- Background work uses Action Scheduler; on quiet or shared hosting, a real server cron job keeps it on time.

13. Getting started

The Square Sync screen guides you through setup. Nothing is pre-filled: you supply your own token, location and categories.

Step	What to do
1. Get a token	In the Square Developer Dashboard, open or create an application, switch to Production, and copy the Production access token.
2. Connect	Paste it on WooCommerce, Square Sync and click Connect to Square. The token is tested first. The webhook is registered, and if your account has exactly one active location it is selected for you.
3. Location	Choose the Square location whose stock the website should use, if it was not picked for you.
4. Categories	Tick Sync on the categories to publish and Skip on any to keep off, then Save settings. Nothing syncs until at least one category is chosen.
5. Options	Optionally set the price-button pattern, new-product status, website order mode, automatic SKU prefix and label shop name.
6. Sync	Click Sync now. Progress shows on the Status card; the site then keeps itself up to date.

Appendix A. Extension points

For developers who want to build on the plugin without editing it.

Hook or option	What it does
Action: dzsq_inventory_count	Fires with the raw Square inventory-count object when a webhook reports a count for a catalog object the plugin did not create. Use it to react to register sales of items that live outside the sync.
Filter: dzsq_product_category_ids	Change which WooCommerce categories a synced product is filed under. Receives the category IDs mapped from Square, the product, the Square item and the existing product ID (0 for a new product).
Option: dzsq_paused	Set to a true value to make the plugin stay completely idle without deactivating it.
Webhook endpoint	/wp-json/dzsq/v1/webhook (POST, signature-verified).
Background jobs	Action Scheduler group dzsq. Queued and recurring jobs are cancelled when the plugin is deactivated.
Request keys	Every Square request key includes a short code derived from the site address, so several websites can share one Square account.

Appendix B. Data the plugin stores

Where	Keys
Products	_dzsq_item_id, _dzsq_version, _dzsq_variation_id, _dzsq_upc, _dzsq_image_ids, _dzsq_keep_hidden, _dzsq_manual_synced
Product categories	_dzsq_category_id (term meta)
Media	_dzsq_image_id (attachment meta)
Orders	_dzsq_square_order_id, _dzsq_square_payment_id, _dzsq_square_total, _dzsq_stock_done, _dzsq_attempts, _dzsq_reversed
Options	dzsq_settings; dzsq_token and dzsq_webhook_key (both encrypted); dzsq_webhook_id; dzsq_webhook_url; dzsq_log; dzsq_last_sku (last SKU number issued); dzsq_label_offset (label calibration); plus short-lived run, lock and status entries
Endpoint and jobs	Webhook route /wp-json/dzsq/v1/webhook; Action Scheduler group dzsq